AMERY ICE SHELF BOREHOLE MICROCAT DATA 2008/2009 SEASON - DATA PROCESSING AND QUALITY

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INTRODUCTION

This report summarises the latest AMISOR microcat data, season 2008/2009, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes. Details of data processing and data quality are included. Temperature/salinity/pressure time series data were downloaded from three strings of Seabird SBE37IM microcats at AMISOR boreholes AM01, AM03 and AM04 (12 to 13 months of new data from each site). At borehole AM01, data were only recoverable from microcat 1969 - batteries were flat and data unrecoverable for microcats 1970 and 1971. For borehole AM02, microcat battery life ended the previous season.

All downloads were done by Mark Milnes, and a complete description of field operations is given in the report:

AMISOR DATA EXTRACTION 2009, Mark Milnes and Roland Stehle, Australian Antarctic Division unpublished report,14pp.

Table 1: Station details for AMISOR season 2008/2009 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values over the data dates.

borehole	position at given date		microcat	mean instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM01	69°24.308'S	71°31.145'E	1969	436.2	441.1	07-Jan-2008	07-Jan-2009	30	+154
AM01	(on 07-Jan-2009)		1970	-	-	-	-	-	-
AM01	"	"	1971	-	-	-	-	-	-
AM02	69°40.949'S	72°43.892'E	1623	-	-	-	-	-	-
AM02	(on 07-Jan-2009)		1624	-	-	-	-	-	-
AM02	"	"	1174	-	-	-	-	-	-
AM03	70°33.065'S	70°20.287'E	4055	652.7	660.4	04-Jan-2008	27-Jan-2009	30	+189
AM03	(on 27-Jan-2009)		4054	860.3	871.0	04-Jan-2008	27-Jan-2009	30	+149
AM03	"	"	3883	1209.6	1225.6	04-Jan-2008	27-Jan-2009	30	+98
AM04	69°53.231'S	70°18.914'E	1972	540.0	546.2	05-Jan-2008	30-Jan-2009	30	+173
AM04	(on 28-Jan-2009)		1973	680.9	689.0	05-Jan-2008	30-Jan-2009	30	+214
AM04	"	"	1974	801.4	811.2	05-Jan-2008	28-Jan-2009	30	+109

MICROCAT DATA PROCESSING

The microcat moorings have already been described in previous reports. The mooring at AM01 was deployed in January 2002; January 2001 for AM02; December 2005 for AM03; and January 2006 for AM04. All latest microcat data downloaded during the 2008/2009 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time (Table 1). Clocks were reset to GPS time for all sets of microcats, following data download and prior to reinitialisation of logging.

Manufacturer supplied calibrations (May/June 2001 for AM01 instruments, June 2005 for AM03 instruments [except October 2004 for microcat-3883 pressure], and February 2003 for AM04 instruments) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* For AM01, pressure data were still bad for the remaining functioning microcat. A constant pressure value was selected for the AM01 instrument (Table 1), from inspection of data near the start of the deployment in January 2002. This pressure value was used to recalculate conductivity, and then in the salinity calculation.

* For AM03 and AM04, a mean pressure value was calculated for the year's data for each microcat (Table 1).

* Decimal times were “compressed” linearly throughout the time series to correct for clock error (Table 1).

MICROCAT DATA QUALITY NOTES

The following bad data were removed from the files:

* AM01, microcat 1969 - bad temperature data removed for the first 2 days; bad conductivity/salinity data removed for the first 23 days

* AM03, microcat 4055 - conductivity/salinity data spikes removed for ~40 data points over the time series

* AM04, microcat 1972 - conductivity/salinity data spikes removed for data points 2699-2721, 2926-2928, 6943

* AM04, microcat 1973 - conductivity/salinity data spike removed for data point 18163

* AM04, microcat 1974 - conductivity/salinity data spikes removed for data points 8533, 18181, 18608-18610

The following data appear suspect (but not removed from the files):

* AM01, microcat 1969 - conductivity/salinity data are suspect up to the start of 27/03/2008 (Figure 1)

* AM01, microcat 1969 - all data for this season appears to be increasingly suspect as the battery on the instrument runs down

Microcat 1969 at AM01 was the only microcat to communicate at this site, and it is likely that the battery will be flat the next time the site is visited. Over the past few seasons, all data for this instrument appear increasingly suspect as the battery flattens. On inspection of the whole data set for this instrument from initial deployment (Figure 5), parameter variance diminishes from ~January 2004 onwards, and the time series are unlike data from any of the other microcats at this and the other boreholes. Data for microcat 1969 from January 2004 onwards should be treated with caution. In particular, data from this latest download should be treated as suspect.

Microcat data from the latest data downloads are plotted in Figures 1 to 4. For both AM03 and AM04, the pressure variation throughout the time series is very similar for all three microcats at each site (Figure 4). Although the order of the drift (~0.5 dbar at AM03 and ~1.5 dbar at AM04) is of the same order as the expected pressure sensor drift as quoted by the manufacturer, it is unlikely that the pressure sensor calibration drift at each site would be so similar for each set of instruments - thus the pressure decrease at AM03 and AM04 over the year appears to be real. For AM01, unfortunately no good pressure data were available from the microcats to observe any pressure changes.

For AM03, numerous strong current events occur between decimal days 2750 and 2900 (Figure 4), causing "push-up" of the instrument string. For AM04, an interesting abrupt "push-up" event occurs around decimal day 2740 (Figure 4) - the close correlation between pressure data from the 3 instruments at this time indicates the event is real. The strong step characteristic of the event hints at a different cause (i.e. not ocean currents) - to speculate, perhaps the cause is local deformation of the ice shelf, effectively lifting the instrument string for a few days. Close inspection of the data shows the instruments steadily lifting over ~3.5 hours at the start of the event, then dropping again within half an hour at the end of the event.

From conductivity and temperature sensor calibration drift estimates in a previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2008/2009 data downloads:

AM01 (in water for 7 years)	total S drift = 0.025 (PSS78)	total T drift = 0.0035^{o}C
AM03 (in water for 3 year)	total S drift = 0.011 (PSS78)	total T drift = 0.0015^{o}C

For AM04, manufacturer sensor calibrations were from February 2003, and the following total drifts might be expected at the time of the latest 2008/2009 data downloads (despite only 3 years in the water):

AM04 (after 6 years)	total S drift = 0.022 (PSS78)	total T drift = 0.0030^{o}C

Given the apparent real pressure changes observed at AM03 and AM04 (Figure 4), similar behaviour might be expected at AM01. So the lack of new pressure data from this site means there is an expected pressure error, which in turn would lead to a salinity error. In the worst case, this salinity error could potentially be up to 0.01 PSS, however without real pressure variation data at the site any estimate of this error is only a guess.

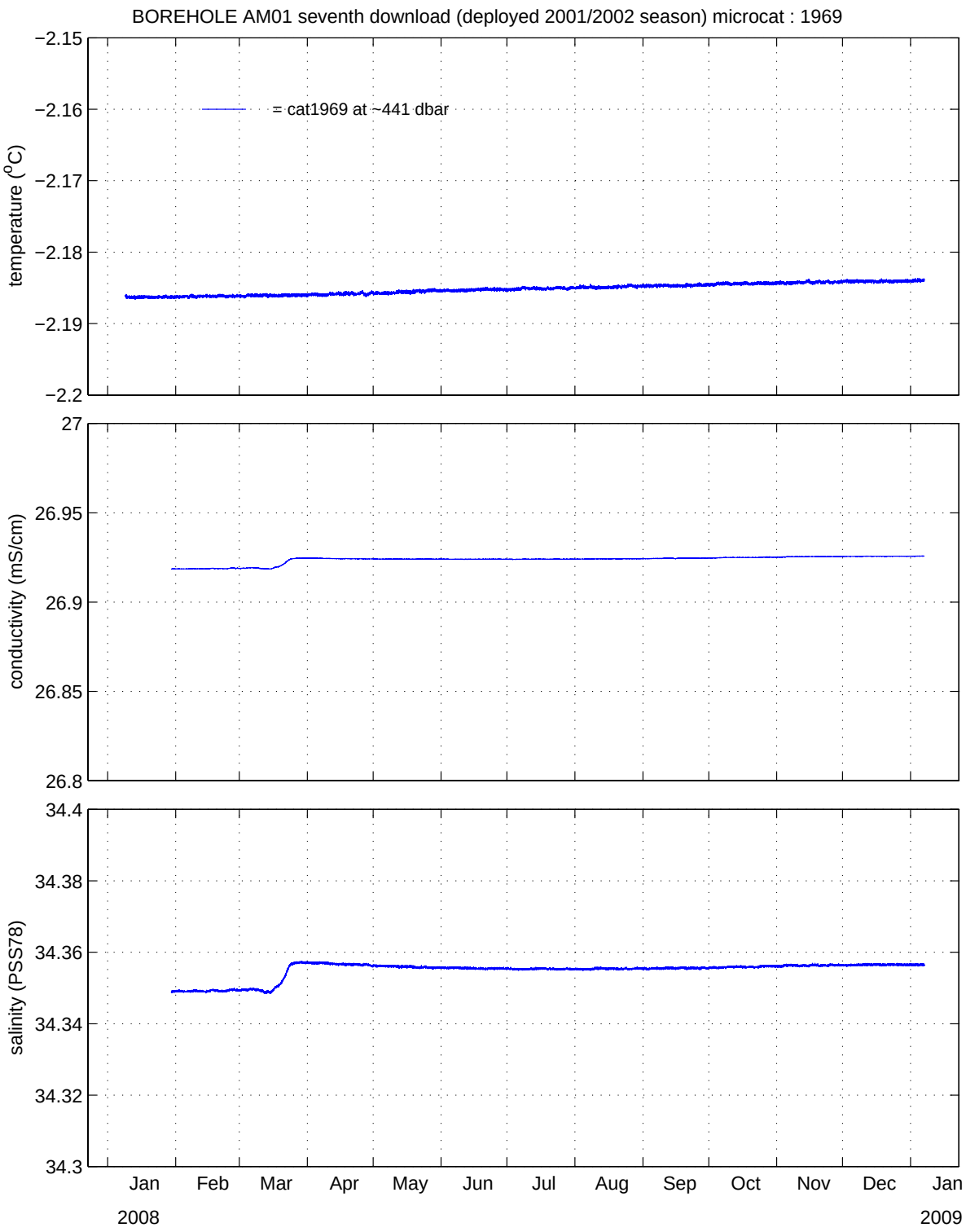


Figure 1: **Borehole AM01 microcat data downloaded in 2008/2009 season.**

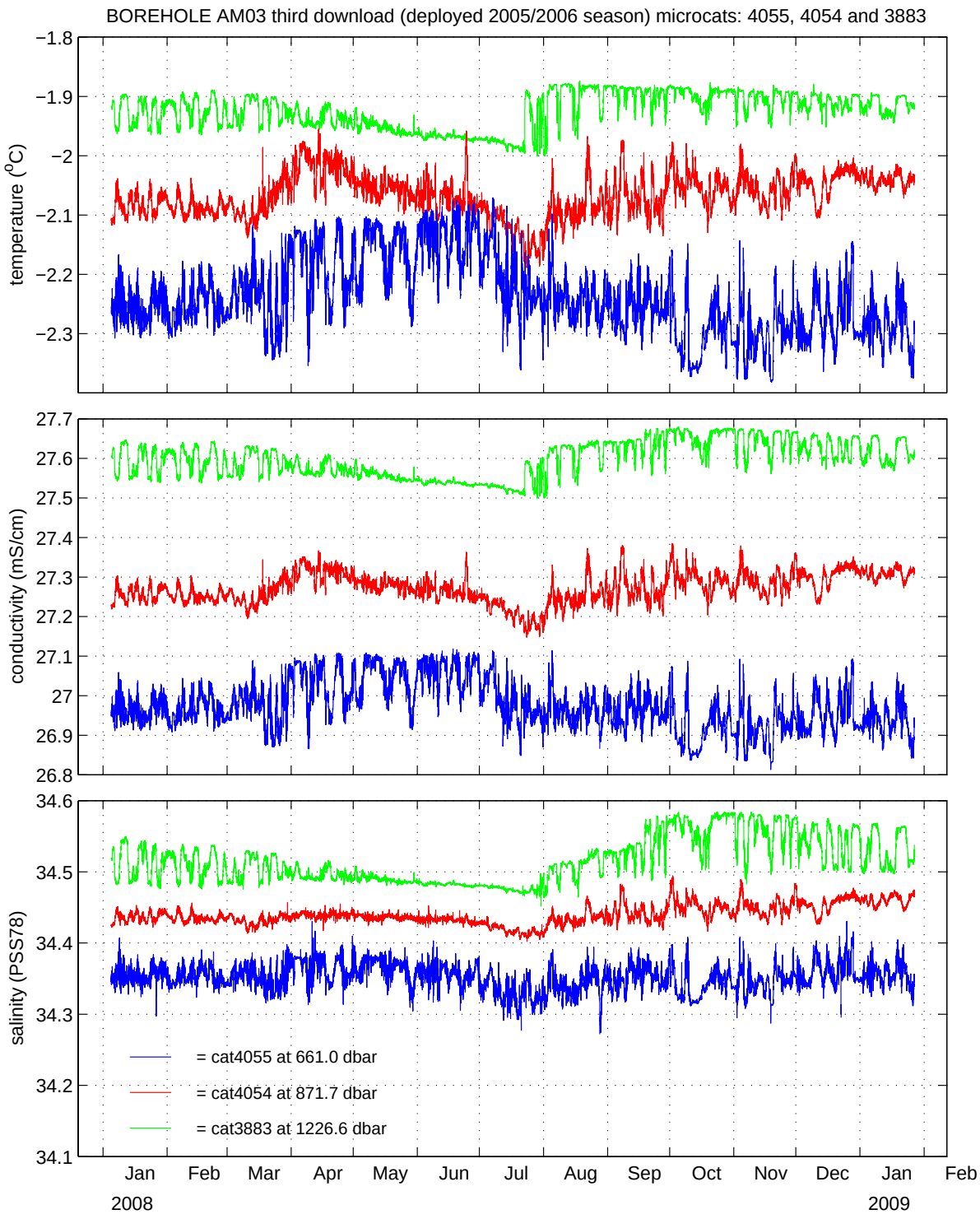


Figure 2: **Borehole AM03 microcat data downloaded in 2008/2009 season.**

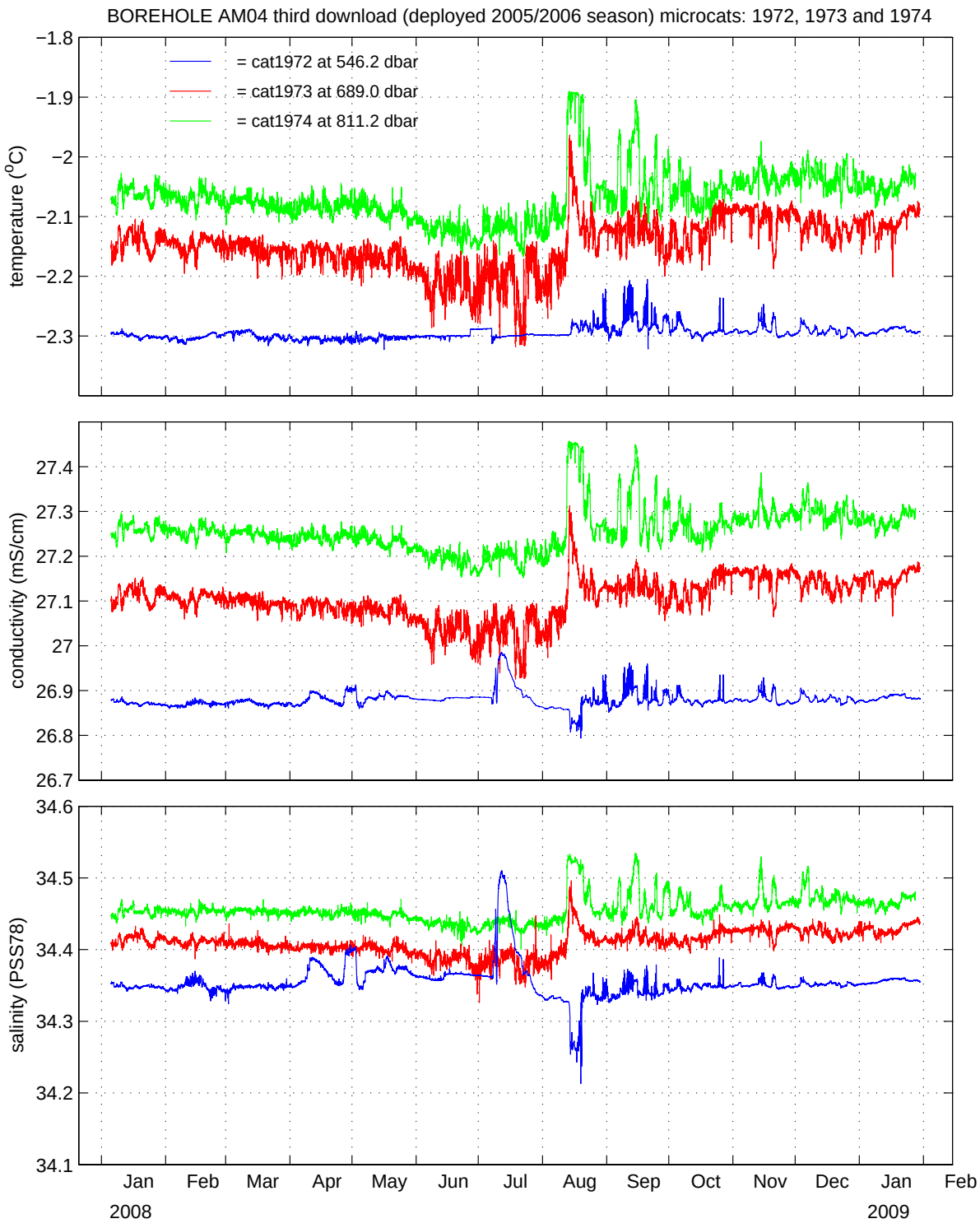
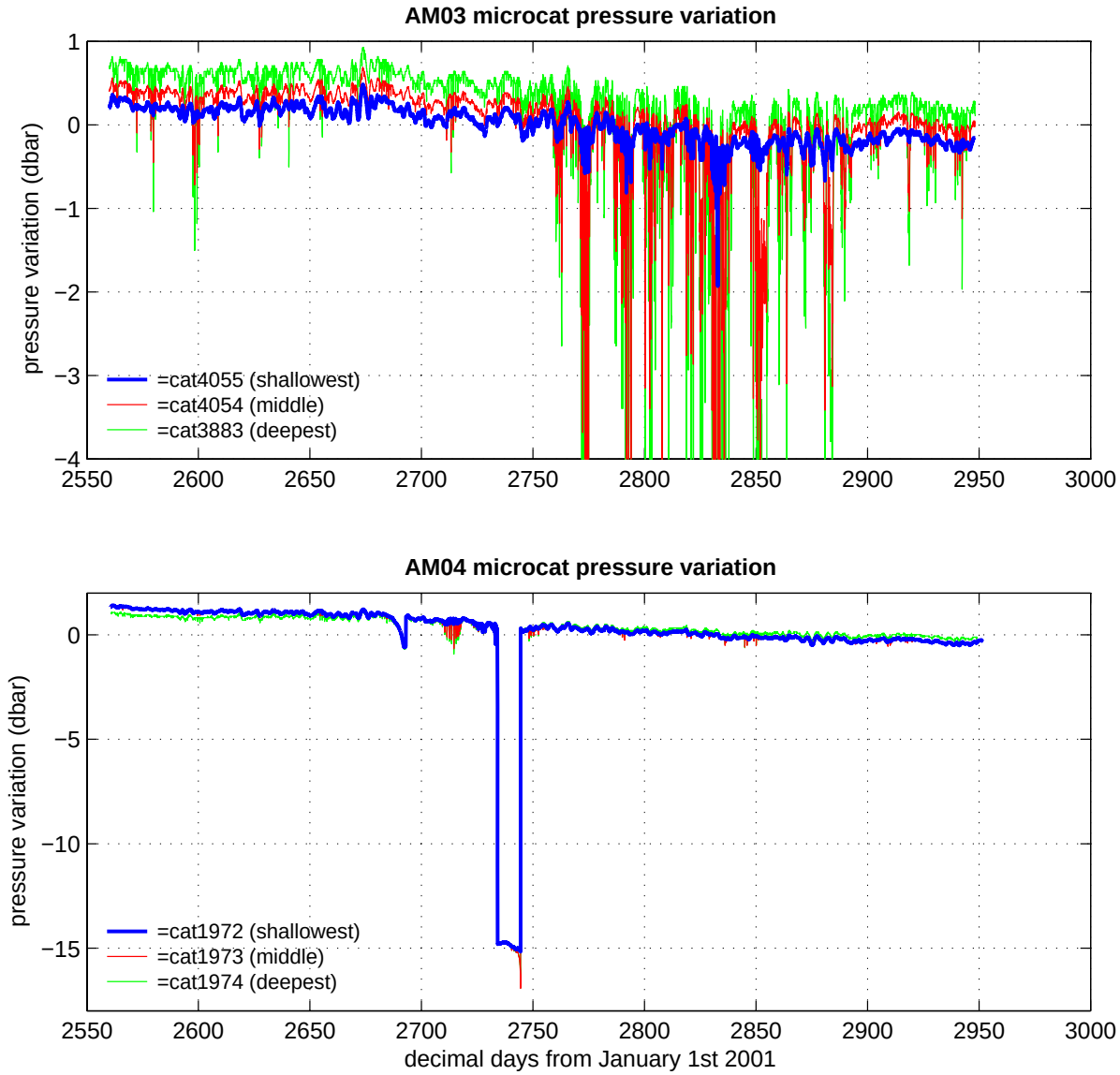
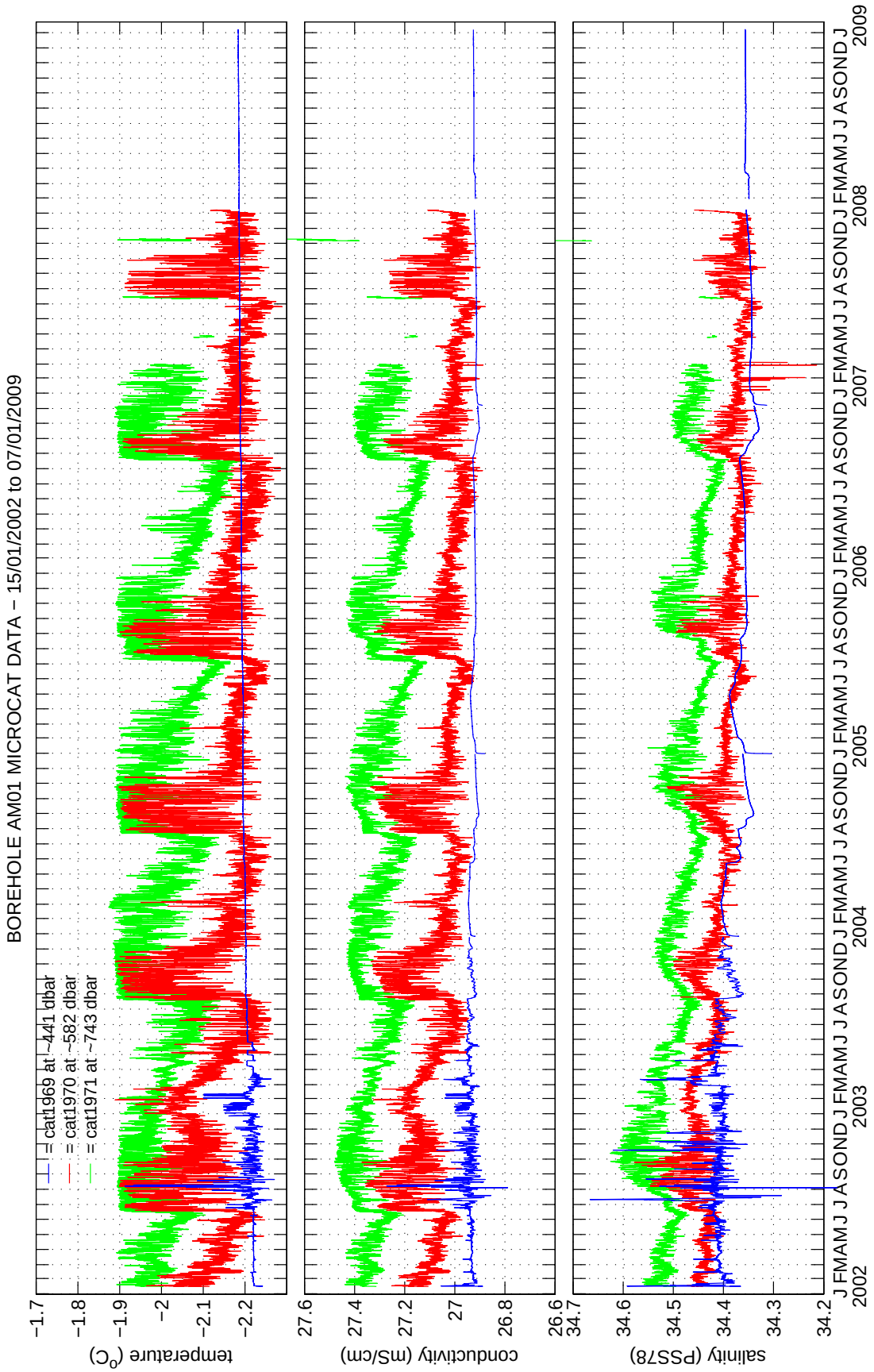


Figure 3: **Borehole AM04 microcat data downloaded in 2008/2009 season.**



<u>Figure 4:</u> Pressure sensor data from boreholes AM03 and AM04 microcat data downloaded in 2008/2009 season. Plotted pressures are variations about the mean values.



<u>Figure 5:</u> Entire time series of borehole AM01 microcat data from initial deployment, including latest data dowloaded in 2008/2009 season.

DATA FILES

1. Microcat data from latest downloads

New microcat data from boreholes AM01, AM03 and AM04 are contained in text and matlab format files, as follows (note that there's no new AM02 data):

AM01
Data from the seventh data download (i.e. 07/01/2008-07/01/2009) are contained in the files:
cat1969_7.dat (text file) and cat1969_7.mat (matlab file).
READMEcatam01_7 details these microcat file formats.

AM03
Data from the third data download (i.e. 04/01/2008-27/01/2009) are contained in the files:
cat4055_3.dat, cat4054_3.dat, cat3883_3.dat (text files), and
cat4055_3.mat, cat4054_3.mat, cat3883_3.mat (matlab files).
READMEcatam03_3 details the microcat file formats.

AM04
Data from the third data download (i.e. 05/01/2008-30/01/2009) are contained in the files:
cat1972_3.dat, cat1973_3.dat, cat1974_3.dat (text files), and
cat1972_3.mat, cat1973_3.mat, cat1974_3.mat (matlab files).
READMEcatam04_3 details the microcat file formats.

2. Unified microcat data files

The updated unified files (i.e. all data from initial deployment time, updated with data from the latest download) are contained in the following files:

AM01 (i.e. 15/01/2002-07/01/2009)
cat1969_all.dat, cat1970_all.dat, cat1971_all.dat (text files), and
cat1969_all.mat, cat1970_all.mat, cat1971_all.mat (matlab files).
READMEcatam01_all details these microcat file formats.
catamisor1all.eps is a time series plot.

AM02 (i.e. 06/01/2001-16/12/2006)
note: no new data here as all batteries are now flat
cat1623_all.dat, cat1624_all.dat, cat1174_all.dat (text files), and
cat1623_all.mat, cat1624_all.mat, cat1174_all.mat (matlab files).
READMEcatam02_all details these microcat file formats.
catamisor2all.eps and catamisor2allp.eps are time series plots.

AM03 (i.e. 27/12/2005-27/01/2009)
cat4055_all.dat, cat4054_all.dat, cat3883_all.dat (text files), and
cat4055_all.mat, cat4054_all.mat, cat3883_all.mat (matlab files).
READMEcatam03_all details these microcat file formats.
catamisor3all.eps and catamisor3allp.eps are time series plots.

AM04 (i.e. 11/01/2006-30/01/2009)
cat1972_all.dat, cat1973_all.dat, cat1974_all.dat (text files), and
cat1972_all.mat, cat1973_all.mat, cat1974_all.mat (matlab files).
READMEcatam04_all details these microcat file formats.
catamisor4all.eps and catamisor4allp.eps are time series plots.

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.